🧠 Brain dump · NRSG 224 Patho · Circulatory 1–2 + Class 8 (Circ 3 / Resp 1)

From your exam notes only · memory tricks are in the app (marked "not from professor") · print 2-sided

Chambers & what fails

Circ 1

Chamber	Job	If it fails
Atria (2, top)	Collect returning blood + pump to fill ventricles	↓CO · blood pools → clots → emboli → ischemia
Right ventricle	→ lungs (pulmonary circ.)	Blood not oxygenated
Left ventricle	→ body (systemic circ.)	Tissues starve → organs fail · ↓CO
Valves	One-way flow	—
Pericardium · Myocardium · Endocardium	Sac/position · muscle work · inner lining	See "Heart layers"

The 2 big equations

Circ 1

Equation	Means	Remember
CO = HR × SV	Blood/min = beats/min × blood/beat	HR too fast (~150–200) → no filling → CO ↓
SV set by	Preload · Afterload · Contractility	Interventions: ↑preload, ↓afterload, ↑contractility
BP = CO × PVR	More CO or smaller vessels → ↑BP	PVR = mostly vessel diameter
Normal BP	120/80	Normal = BOTH <120 AND <80

Preload · Afterload · Contractility

Circ 1

	What	↑ does…	Limit / example
Preload	Blood in ventricle ready for next beat (venous return)	More stretch → ↑SV (Frank-Starling)	Max at ~2½× resting length; overstretch → ↓SV
Afterload	Resistance to ejection (arterial BP)	↑BP → ↓SV → ↓CO	Bike-tire analogy
Contractility	Force independent of pre/afterload	+ inotrope ↑ (sympathetic) · − inotrope ↓	− : hypoxia, MI

BP too low / too high

Circ 1

Problem	Fix	Why (equation)
BP LOW	IV fluid	↑volume
BP LOW	Vasoconstrictor	↑PVR
BP HIGH	Diuretic	↓volume
BP HIGH	Slow HR drug	↓CO
BP HIGH	Vasodilator	↓PVR

Controllers & RAAS

Circ 1

Controller	Affects	Notes
Autonomic NS	HR + vessel diameter (NOT volume)	Medulla: vasomotor (symp.) · cardioinhibitory (parasymp.)
Kidneys	Volume (urine) + RAAS	Low BP → renin → Ang I → (lung) Ang II → vasoconstrict + aldosterone → Na/H2O held
ADH	Volume + vessels	Posterior pituitary; vasoconstricts, holds water

Magic numbers

All 3

Thing	Number
Normal BP	<120 AND <80 (≈120/80)
Cholesterol	Total <200 · LDL <100 · TG <200 · HDL >40 (≥60 protective)
Alcohol (excess)	>2/day men · >1/day women
NTG	3 doses × 5 min apart → then think MI
Thrombolytic (MI)	Start within 12 h · best within 2 h
CABG grafts	Often 3–4
HR limit for filling	~150–180–200 (fitness)
Starling limit	~2½× resting length
SA node · AV node	60–100 · 40–60 bpm
Brady · Tachy	<60 · >100
V-tach	100–250/min
Resp failure	PO2 <50 and/or PCO2 >50 (normal PO2 >80–100; PCO2 35–45)
O2 carried	~1% dissolved · 98–99% on Hgb
CO2 carried	~10% dissolved · ~30% Hgb · ~60% bicarbonate
Hgb releases O2	~PO2 60 normal · ~80 (↓affinity) · holds to ~40 (↑affinity)
Incentive spirometer	10× every hour while awake

Angina types

Circ 1

Type	Cause	Pain	Relief
Stable (classic)	Demand ↑ (exertion, emotion, cold, big meal)	Predictable	Rest / NTG
Variant (Prinzmetal)	Coronary SPASM	Often at rest, night/early AM	—
Unstable (crescendo)	Unstable plaque	Rest/minimal exertion; longer, more frequent, new	NOT reliably · MI warning

MI toolkit

Circ 1

Treatment	What it does	Key point
Nitroglycerin	Vasodilates arteries (↓afterload) + veins (↓preload)	↓demand, NOT ↑supply · 3×5
Aspirin (low-dose)	Antiplatelet	Avoid w/ gastric ulcers, aneurysms
Thrombolytic (MI only)	BREAKS clots, saves injured cells	≤12 h, best ≤2 h; bleeding risk = no
Anticoagulant (heparin, Coumadin)	Stops growth/new clots	Cannot break a clot
Morphine (MI only)	Pain relief	Pain → sympathetic → tachycardia → ↑workload
PTCA · CABG	Balloon opens · bypass	Revascularization

Heart valves

Circ 2

Type	Valve	From → To
AV	Tricuspid	R atrium → R ventricle
AV	Mitral (bicuspid)	L atrium → L ventricle
Semilunar	Pulmonic	R ventricle → pulmonary artery
Semilunar	Aortic	L ventricle → aorta
Stenosis	Doesn't open fully	Chamber behind hypertrophies, can fail
Regurgitation	Doesn't close fully	Backflow → receiving chamber volume-overloads

Heart layers

Circ 2

Layer	Function	Problem
Pericardium	Position · protect · barrier · prevents overfilling	Effusion → tamponade (compresses heart → CO ↓, emergency)
Myocardium	Muscle does the work (involuntary)	Cardiomyopathy → ↓SV/CO
Endocardium	Inner lining	Infective endocarditis: vegetations (bacteria+fibrin+platelets), may embolize

Right vs Left heart failure

Circ 2

	RIGHT HF	LEFT HF
Forward	↓blood to lungs → ↓systemic CO	↓arterial flow → low perfusion → cardiogenic shock
Backward goes to…	BODY (venous)	LUNGS
Signs	JVD · dependent edema · hepatomegaly · splenomegaly · ascites	Pulmonary edema · crackles · orthopnea · PND

HF: types, compensation, drugs

Circ 2

Topic	Key points
HFrEF	Reduced EF = systolic = doesn't EMPTY
HFpEF	Preserved EF = diastolic = doesn't FILL
Compensations	↑Sympathetic · RAAS · remodeling (hypertrophy)
ANP/BNP	Released on stretch → ↓SNS, ↓Na/H2O retention, ↓vasoconstriction · BNP = severity marker
Drug goals	↑contractility · ↓edema · ↓workload · counteract SNS + RAAS

Arterial vs Venous (acute vs chronic)

Circ 2

	Arterial	Venous
Chronic	PAD: claudication, cool/pale, weak pulses, hair loss, shiny skin, eschar	CVI: dependent edema, brown skin, medial-ankle ulcers, aching standing
Acute	Occlusion: 7 P's (Pallor, Polar, Pulselessness, Pain, Paresthesias, Paralysis, Pistol shot)	DVT: Virchow = stasis · hypercoag. · wall injury → PE risk
Feels	Cool, pale; pain w/ elevation/walking	Warm, brown, edema; pain w/ dependency

Conduction & refractory periods

Class 8

Piece	Role	Detail
SA node	Primary pacemaker	60–100 bpm
AV node	Delays impulse (atria empty)	Backup 40–60
His → branches → Purkinje	Fast → ventricles contract together	Max CO
Absolute refractory	No new AP, ever	Longer in heart → no sustained contraction
Relative refractory	Needs bigger stimulus	Abnormal → dysrhythmias
Supernormal	Weak stimulus works	Abnormal → dysrhythmias

EKG & dysrhythmias

Class 8

Item	What	CO effect
P · QRS · T	Atria depol. · ventricles depol. · ventricles repol.	—
Normal sinus rhythm	P before every QRS + even R-R	—
PVC	Wide, bizarre QRS (ectopic)	↓filling
Brady <60 / Tachy >100	Rate problems	Both can ↓CO
A-fib	Disorganized atria	Pools → thrombus → emboli; no atrial kick
Heart block	1° delayed PR · 2° intermittent · 3° independent	↓CO
V-tach	100–250/min	↓filling, no atrial kick
V-fib	Quivering, no QRS	CO ≈ 0 → DEFIBRILLATE

Shunt vs Dead space

Class 8

	Shunt	Dead air space
Definition	Perfused, NOT ventilated	Ventilated, NOT perfused
Causes	Alveolar collapse, obstruction, ↓chest expansion, depressed resp center	Pulmonary embolism (classic)
Her example	Something occluding the airway	Obstruction in pulmonary capillary
Diffusion needs	Pressure gradient · surface area · membrane thickness	

Compliance & airways

Class 8

Topic	Points
↓ Compliance (stiff)	Fibrosis · pulmonary edema · ARDS · atelectasis · ↓surfactant
↑ Compliance (too floppy)	Emphysema · aging (easy in, hard out)
Airway resistance	Narrow → ↑work of breathing
Surfactant (Type II)	↓surface tension; lack → atelectasis → TCDB + IS
Pleural space	Negative pressure; air in → pneumothorax → chest tube

Hemoglobin affinity & gas transport

Class 8

	↓ Affinity (releases O2)	↑ Affinity (holds O2)
Conditions	More acid · CO2 · heat	Less acid · CO2 · heat
Metabolism	↑	↓
Release point	~PO2 80	Holds to ~PO2 40
Normal	Starts releasing ~PO2 60	

Hypoxia, CO2 & control

Class 8

Topic	Points
Hypoxia	Low O2 to tissues; SNS first (tachy, pale/clammy, anxious) → brain; cyanosis LATE
Ischemia	Low BLOOD to tissues
Cap refill	Checks circulation, not O2
Hypercapnia	CO2 narcosis: sleepy → disoriented → coma
Resp failure	PO2 <50 and/or PCO2 >50; look at the patient
Control	Pons + medulla · chemoreceptors (medulla, carotids, aorta: O2/CO2/pH) · cortex voluntary
ANS	Sympathetic → bronchodilation · Parasympathetic → bronchoconstriction

Source: your NRSG 224 exam-notes docs (Circulatory Part 1, Circulatory Part 2, Class 8). Study aid only — graded work stays your own.